

Hyperbolic Function Cheat Sheet

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\tanh(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$\coth(x) = \frac{e^{2x} + 1}{e^{2x} - 1}$$

$$\operatorname{sech}(x) = \frac{2}{e^x + e^{-x}}$$

$$\operatorname{csch}(x) = \frac{2}{e^x - e^{-x}}$$

$$\operatorname{arcsinh}(x) = \ln \left(x + \sqrt{x^2 + 1} \right)$$

$$\operatorname{arccosh}(x) = \ln \left(x + \sqrt{x^2 - 1} \right), \quad x \geq 1$$

$$\operatorname{arctanh}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad |x| < 1$$

$$\operatorname{arcoth}(x) = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right), \quad |x| > 1$$

$$\operatorname{arcsech}(x) = \ln \left(\frac{1}{x} + \frac{\sqrt{1-x^2}}{x} \right), \quad 0 < x \leq 1$$

$$\operatorname{arccsch}(x) = \ln \left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|} \right), \quad x \neq 0$$

$$\sinh \theta \operatorname{csch} \theta = 1$$

$$\cosh \theta \operatorname{sech} \theta = 1$$

$$\tanh \theta \coth \theta = 1$$

$$\tanh \theta \cosh \theta = \sinh \theta$$

$$\sinh \theta \coth \theta = \cosh \theta$$

$$\sinh \theta \operatorname{sech} \theta = \tanh \theta$$

$$\cosh \theta \operatorname{csch} \theta = \coth \theta$$

$$\tanh \theta \operatorname{csch} \theta = \operatorname{sech} \theta$$

$$\coth \theta \operatorname{sech} \theta = \operatorname{csch} \theta$$

$$\sinh^2 \theta + 1 = \cosh^2 \theta$$

$$\tanh^2 \theta + \operatorname{sech}^2 \theta = 1$$

$$\coth^2 \theta + \operatorname{csch}^2 \theta = 1$$

$$\cosh x + \sinh x = e^x$$

$$\cosh x - \sinh x = e^{-x}$$

$$\sinh \theta = \frac{2 \tanh \frac{\theta}{2}}{1 - \tanh^2 \frac{\theta}{2}}$$

$$\cosh \theta = \frac{1 + \tanh^2 \frac{\theta}{2}}{1 - \tanh^2 \frac{\theta}{2}}$$

$$\tanh \theta = \frac{2 \tanh \frac{\theta}{2}}{1 + \tanh^2 \frac{\theta}{2}}$$

$$\sinh 2\theta = 2 \sinh \theta \cosh \theta$$

$$\begin{aligned} \cosh 2\theta &= 1 + 2 \sinh^2 \theta \\ &= 2 \cosh^2 \theta - 1 \\ &= \cosh^2 \theta + \sinh^2 \theta \end{aligned}$$

$$\sinh \frac{\theta}{2} = \operatorname{sgn}(\theta) \sqrt{\frac{\cosh \theta - 1}{2}}$$

$$\cosh \frac{\theta}{2} = \sqrt{\frac{\cosh \theta + 1}{2}}$$

$$\begin{aligned} \tanh \frac{\theta}{2} &= \operatorname{sgn}(\theta) \sqrt{\frac{\cosh \theta - 1}{\cosh \theta + 1}} \\ &= \frac{\sinh \theta}{\cosh \theta + 1} \\ &= \frac{\cosh \theta - 1}{\sinh \theta} \\ &= \frac{\sinh \theta + \cosh \theta - 1}{\sinh \theta + \cosh \theta + 1} \\ &= \coth \theta - \operatorname{csch} \theta \end{aligned}$$

$$\sinh 3\theta = 3 \sinh \theta + 4 \sinh^3 \theta$$

$$\cosh 3\theta = 4 \cosh^3 \theta - 3 \cosh \theta$$

$$\tanh 3\theta = \frac{3 \tanh \theta + \tanh^3 \theta}{1 + 3 \tanh^2 \theta}$$

$$\sinh(\alpha + \beta) = \sinh \alpha \cosh \beta + \cosh \alpha \sinh \beta$$

$$\sinh(\alpha - \beta) = \sinh \alpha \cosh \beta - \cosh \alpha \sinh \beta$$

$$\cosh(\alpha + \beta) = \cosh \alpha \cosh \beta + \sinh \alpha \sinh \beta$$

$$\cosh(\alpha - \beta) = \cosh \alpha \cosh \beta - \sinh \alpha \sinh \beta$$

$$\tanh(\alpha + \beta) = \frac{\tanh \alpha + \tanh \beta}{1 + \tanh \alpha \tanh \beta}$$

$$\tanh(\alpha - \beta) = \frac{\tanh \alpha - \tanh \beta}{1 - \tanh \alpha \tanh \beta}$$

$$\coth(\alpha + \beta) = \frac{\coth \alpha \coth \beta + 1}{\coth \alpha + \coth \beta}$$

$$\coth(\alpha - \beta) = \frac{\coth \alpha \coth \beta - 1}{\coth \beta - \coth \alpha}$$

$$\operatorname{sech}(\alpha + \beta) = \frac{\operatorname{sech} \alpha \operatorname{sech} \beta}{1 + \tanh \alpha \tanh \beta} = \frac{\operatorname{csch} \alpha \operatorname{csch} \beta}{\coth \alpha \coth \beta + 1}$$

$$\operatorname{sech}(\alpha - \beta) = \frac{\operatorname{sech} \alpha \operatorname{sech} \beta}{1 - \tanh \alpha \tanh \beta} = \frac{\operatorname{csch} \alpha \operatorname{csch} \beta}{\coth \alpha \coth \beta - 1}$$

$$\operatorname{csch}(\alpha + \beta) = \frac{\operatorname{csch} \alpha \operatorname{csch} \beta}{\coth \alpha + \coth \beta} = \frac{\operatorname{sech} \alpha \operatorname{sech} \beta}{\tanh \alpha + \tanh \beta}$$

$$\operatorname{csch}(\alpha - \beta) = \frac{\operatorname{csch} \alpha \operatorname{csch} \beta}{\coth \beta - \coth \alpha} = \frac{\operatorname{sech} \alpha \operatorname{sech} \beta}{\tanh \alpha - \tanh \beta}$$

$$\sinh^2 \theta = \frac{\tanh^2 \theta}{1 - \tanh^2 \theta}$$

$$\cosh^2 \theta = \frac{1}{1 - \tanh^2 \theta}$$

$$\coth^2 \theta = \frac{1}{\tanh^2 \theta}$$

$$\operatorname{sech}^2 \theta = 1 - \tanh^2 \theta$$

$$\operatorname{csch}^2 \theta = \frac{1 - \tanh^2 \theta}{\tanh^2 \theta}$$

$$\sinh \alpha + \sinh \beta = 2 \sinh \frac{\alpha + \beta}{2} \cosh \frac{\alpha - \beta}{2}$$

$$\sinh \alpha - \sinh \beta = 2 \cosh \frac{\alpha + \beta}{2} \sinh \frac{\alpha - \beta}{2}$$

$$\cosh \alpha + \cosh \beta = 2 \cosh \frac{\alpha + \beta}{2} \cosh \frac{\alpha - \beta}{2}$$

$$\cosh \alpha - \cosh \beta = 2 \sinh \frac{\alpha + \beta}{2} \sinh \frac{\alpha - \beta}{2}$$

$$2 \sinh \alpha \cosh \beta = \sinh(\alpha + \beta) + \sinh(\alpha - \beta)$$

$$2 \cosh \alpha \sinh \beta = \sinh(\alpha + \beta) - \sinh(\alpha - \beta)$$

$$2 \cosh \alpha \cosh \beta = \cosh(\alpha + \beta) + \cosh(\alpha - \beta)$$

$$2 \sinh \alpha \sinh \beta = \cosh(\alpha + \beta) - \cosh(\alpha - \beta)$$